

Ethiopian Language Support for the **Babel** Package

Version 0.7

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Abstract

The Ethiopian script differs considerably from the Latin script. Most important, it consists of more than 350 different letters. A new transcription method is presented that can be used for the $\text{\LaTeX}$ typesetting system. It was implemented on the basis of the multilingual typesetting package `babel`.

In addition to a guide to the concrete usage of our system, we provide technical details of the implementation and sketch the reasons for our design decisions. Linguistic and historical information on the Ethiopian script is also included.

አገጽፎተ ጽሁፍ

የኢትዮጵያ ጽሁፍ ከላቲን ጽሁፍ እጅግ የተለየ ነው። ጎልቶ የሚታየው ከ 350 በላይ ፊደላትን የያዘ መሆኑ ነው። እነዚህን ፊደላት $\text{\LaTeX}$ በተባለ የጽሁፊት አጣጣል ማቀነባበሪያ ዘዴ የኢትዮጵያን ጽሁፍ በላቲን ቋንቋ ለመጻፍ የሚያስችል አዲስ ያላጻጻፍ ዘዴ (Transcription) ተዘጋጅቶ ቀርቧል።

የኛን የላጻጻፍ ዘዴ ለመጠቀም ከሚያስችል መመሪያ ጋር፤ የአጠቃቀሙን ቴክኒካል ዝርዝርና ለምን ይሄን የላጻጻፍ መንገድ ለመጠቀም እንደወሰነ አሳይተናል። ከላ ኢትዮጵያ ቋንቋዎች ላጻጻፍ ዘዴና የቋንቋ ታሪካዊ መረጃዎች በዚህ ጥናት ላይ አካላንበታል።

1 Introduction

The `ethiop` package [3] is a collection of fonts and $\TeX$ macros that enable you to typeset the characters of the languages of Ethiopia.

1.1 The origins of the Ethiopian script

The Ethiopian script has its origin in the South Semitic alphabet which has been used for Thamudene, Dedanite, Liḥyanite, Ṣafaitic, Minaean, Sabaeen, Ḥimyaritic, Qatabanic, and Ḥaḍramautic. These are Semitic languages once spoken in the present state of Yemen. The alphabet was without vocalization. The South Semitic characters are known from stone monuments only, whereas the present Ethiopian characters originate from paper documents. The writing direction of South Semitic was from right to left or alternating (Bustrofedon).

In the 2nd half of the 1st millennium BC Semitic groups (Ge'ez, ጊዕዝ) from South Arabia established a kingdom at Aksum. After 350 AD a vocalization was introduced.

Today the Ethiopian script is an official writing system in two states, Ethiopia (ኢትዮጵያ) and Eritrea (ኢርትራ). There exist 86 languages (4 of them extinct) from 4 language families with around 200 dialects in Ethiopia and Eritrea. The writing system is used for a number of languages in these states, see Tab. 1 for the most important languages (figures taken from [6]). Note that Amarinya, Guraginya, Afarinya, Hadiyyinya, Kambaatinya, Orominya (Oromiffa), Sidaminya, Somalinya and some others are the expressions in Amharic for the different languages whereas the names in the second column of Tab. 1 are from [6].

1.2 The origins of this package

The current Ethiopian $\LaTeX$ environment started as a project assignment at the university of Hamburg in 1995. Luckily, we did not need to start from scratch, because there were some Ethiopian fonts already.

The fonts are based on Eth $\TeX$ which was originally distributed by Abass B. Alameneh. The genuine Eth $\TeX$ package can be found on CTAN in the directory `language/ethiopia/ethtex/` or in [5]. We changed the existing letters a little to make them more robust at low resolutions, but the most effort went into adding new characters that were not present in the original fonts. Still more important is the addition of $\TeX$ -ligatures to the fonts in order to support our input transcription.

When we chose our transcription scheme, it became apparent that we would have to activate some of the input characters. Although we wrote our own set of macros to handle the activation, we soon decided not to introduce yet another incompatible mechanism for this task. Instead we used the `babel` package by Johannes L. Braams [4] as framework for the implementation of the input transcription. The $\TeX$ macros are derived from the file `language.skeleton` provided with that package, which also allows a more well-rounded support of multiple languages.

Since the original Eth $\TeX$ package used a special editor that is not available for all platforms, it was not easily portable. Moreover, a special version of $\LaTeX$ 2.09 was generated, thereby preventing an upgrade to the now current

Language family	Languages	No. of speakers in 1000 (year 91–95)
Semitic	Ge'ez ግዕዝ	only in church
	Amharic አማርኛ	20 000
	Gurage ኅራግኛ	1 850
	Tigre ትግረ	?
	Tigrinya ትግርኛ	6 050
Cushitic	Afar አፋርኛ	750
	Hadiyya ሀዲያኛ	1 000
	Kambaata ከምባትኛ	1 000
	Oromo ኦሮምኛ	14 000
	Sidamo ሲዳሞኛ	1 500
Omotic	Somali ሶማልኛ	2 050
	Gamo/Gofa/Dawro ጋምኛ	780
East Sudanic	Wolaytta ወላይትኛ	2 000
		?

Table 1: Languages and number of speakers

L^AT_EX2_ε. By rewriting the language support completely and by embedding the Ethiopian fonts into the framework of the `babel` package we got a more robust and portable system that will be usable with future L^AT_EX versions.

Our transcription method does not provide support for the direct entry of Ethiopian characters. Instead a natural encoding has been developed that allows us to enter Ethiopian text via Latin letters. This encoding is based on scientific transcription techniques and is closely related to other encoding standards. One of these standards is SERA, which is mainly intended for the recording and transmission of Ethiopian text within an ASCII environment. However, the SERA encoding was not realizable as an input encoding for T_EX.

Unicode, too, is an important text format, which provides a unified framework for *all* languages by encoding characters with 16 bits instead of 8. The T_EX extension Ω can handle Unicode input. A rudimentary support for Ω has already been included in this package. Note that the Ethiopian script is not a full part of Unicode. Although it has been approved by the Unicode committee and has passed the ISO/JTC1/SC2 ballot, it still awaits the ISO/JTC1 ballot and the final publication.

2 Installation

Installing the `ethiop` package is not overly difficult, especially if you adhere to the following instructions.

1. Check the prerequisites for this package.
 - Make sure you have installed T_EX and METAFONT.
 - Make sure that the files `cmbase.mf` and `romand.mf` from the Computer Modern fonts are accessible to METAFONT.

8. Remove the font files that might be left over from a previous version of ethiop, that is all files `eth*.gf` and `eth*.pk`.
9. Verify the installation by generating this documentation file from its source `ethiodoc.tex`. Run `LATEX` twice to get the references right.

```
latex ethiodoc.tex
latex ethiodoc.tex
```

The resulting file `ethiodoc.dvi` should be identical to this text except for the date of translation.

It is highly probable that after doing all of the above, you have a working Ethiopian language package set up. But maybe you ran into trouble during the installation. In this case try the following:

- If the run on `ethiop.ins` fails, the files might have been corrupted during transmission or one of the necessary files might not be accessible to `TEX`.
- If `LATEX` complains about a missing input file, check whether the files `*.fd` and `*.sty` are accessible and readable.
- If `LATEX` complains about a missing font, check the placement of the files `*.tfm`.
- If `LATEX` issues warning messages, you might have an old version of `LATEX` or `babel`.
- If `LATEX` issues strange errors, you might have an old version of `LATEX` or `babel`.
- If `LATEX` issues strange errors, the files might have been corrupted during transmission. Conversions of CR, LF, and so on might cause this problem.
- If the previewer or the printer driver complains about missing fonts and does not automatically call METAFONT to generate these font, either adapt your installation or generate the fonts by hand. Depending on your installation you must run something like

```
mf '\mode=localfont; input etha10'
```

for each of the fonts.

- If METAFONT is called and complains about missing source files, check whether these (`*.mf`) are placed correctly.
- If METAFONT is called and complains about missing source files, check whether the Computer Modern fonts have been properly installed.
- If METAFONT is called and complains about strange paths, you are probably generating the font at a lower resolution than 200 dpi. There is little you can do except ignoring the errors or telling METAFONT to do so. Please report such errors only if they occur at resolutions above 200 dpi.

- If the previewer or the printer driver complains about missing characters, check whether you have deleted all files from previous versions of this font.
- If the previewer or the printer driver complains about a checksum error, check whether you have deleted all files from previous versions of this font.

If that does not help, have a look at our web page located at http://www.informatik.uni-hamburg.de/TGI/mitarbeiter/wimis/kummer/ethiop_eng.html to read about possible updates and bug reports. If this does not provide any clue and your friendly $\TeX$ nician is unavailable, we will try to help you, if time permits. Send a mail to ethiop@informatik.uni-hamburg.de, hopefully we will get back to you.

Please do **not** report bugs to Johannes L. Braams (the maintainer of `babel`) or to the $\LaTeX$ team. They are **not** responsible for our bugs and they are **very** busy already.

3 Usage

An important fact about the Ethiopian script is that it uses more than 350 different characters. Hence the text entry itself is a big problem. But we must also deal with other topics like punctuation and spacing that arise when using this package.

We will describe here the use of the `ethiop` package without the Ω system. The Ω support can be found in section 4.3.

3.1 Including the package

The `babel` language support is selected by adding the line

```
\usepackage[english]{babel}
```

to the preamble of a document. Of course a different language might also be selected. But since the `ethiop` package is not yet a part of the `babel` package, you must select it with

```
\usepackage{ethiop}
```

in the preamble of a document. There is no need to include `babel` explicitly unless we want to use two or more languages. If both `ethiop` and `babel` are included, we can use the ordinary language switching mechanism of `babel` to take us from one language to the other. For example

```
\selectlanguage{english}'adis 'ababA
\selectlanguage{ethiop}'adis 'ababA
\selectlanguage{english}'adis 'ababA.
```

will give you: 'adis 'ababA አብሳ አበበ 'adis 'ababA.

3.2 Typing the text

In Tab. 2 you can see the Ethiopian characters that are accessible using this package.

All characters with `.d` are only used for *Orominya (Oromiffa)*, all with `'q`, `'k`, `'h`, `'g` and `mua`, `mui`, `muE`, `mue` only for *Guraginya (Chaha)*, all with `.q` only for *Tigrinya*, and all with `fu`, `pu` only for *Wolayttinya*.

We will now explain how the characters are entered. Every character represents a syllable that consists of a consonant followed by a vowel. If possible, every character is encoded in a way that matches its pronunciation or its scientific transcription as closely as possible.

As an example we choose the letter **ደ** which is pronounced *da* and entered as **da**. You will find the character in the character table in row **d** and column **a**.

The letter **ደ** might represent the syllable *de*, but it might also denote the consonant *d* without an accompanying vowel. To reflect this, it is possible to enter either **de** or **d** at the users choice.

If accented characters are used in the scientific transcription of an Ethiopian syllable, they are entered without the accent, but prefixed with an appropriate special character. **ሸ** has *ča* as its transcription, hence we will enter it as **^ca** which is as close to the proper transcription as we can get.

Long vowels are usually indicated by a bar, *dē* can serve as an example. But since long vowels are fairly frequent, we do not want to use two letters for their representation. Instead we will employ the uppercase letters for this purpose, which leads us to **dE** as our transcription of the aforementioned syllable **ደ**.

When a vowel does not have both a short and a long form, like the *i* in *di*, there is no need to insist on the proper case for the vowel. Hence we might enter either **di** or **dI** and get **ደ** in both cases.

Some consonants may be followed by a diphthong, i.e. a combination of two vowels. A diphthong occurs e.g. in **ደ**, which is best transcribed *dwā*. We will code it as **duA**, thereby slightly deviating from the standard transcription. But this is unavoidable, because if we enter **dwA**, we will get an ambiguity with **dwA** which we want to result in **ደዋ**. The SERA encoding, which is developed by Daniel Yacob, Yitna Firdiyewek, and Yonas Fisseha, suggests **dWA**, which has been considered as an alternative to the present encoding. It did not show any significant advantages, however.

For the consonant series **ፈ ፊ ፋ ፈ ፊ ፋ ፈ ፊ ፋ**, there exists a variant form that looks like **ፈ ፊ ፋ ፈ ፊ ፋ ፈ ፊ ፋ**, and denotes the same syllables. The variant series can be activated by issuing the command `\ethvariantCtrue` once. The original letterforms can then be restored by typing `\ethvariantCfalse`. This option will only be needed in comparative studies, usually the standard series should be preferred.

In the lower right corner of Tab. 2 you can see an inlay with the three characters **ሸ**, **ሹ**, and **ሺ**. These characters are probably the remnants of three complete series of seven syllables that were once used.

We added these three in order to completely cover the character set of the proposed Unicode standard for code positions 1200h to 137Fh. The encoding of these characters is not fixed yet and may change at any time in the future. At the moment the encoding is based on the characters' appearance, but we are aware that the pronunciation is different today.

		a	u	i	ā	ē	e	o	wa	wi	wā	wē	we
		a	u	i	Ā	Ē	e	o	ua	ui	uĀ	uĒ	ue
		U	I					0	uI	<i>if preferred</i>			
h	h	ሀ	ሁ	ሂ	ሃ	ሄ	ህ	ሆ			ሸ		
l	l	ለ	ሉ	ሊ	ላ	ሌ	ል	ሎ			ሸ		
h	.h	ሐ	ሑ	ሒ	ሓ	ሔ	ሐ	ሑ			ሐ		
m	m	መ	ሙ	ሚ	ማ	ሚ	ም	ም	ጠጐ	ጠጐ	ሚ	ማ	ምጐ
ś	's	ሠ	ሡ	ሢ	ሣ	ሤ	ሥ	ሦ			ሢ		
r	r	ረ	ሩ	ሪ	ራ	ሪ	ር	ሮ			ሪ		
s	s	ሰ	ሱ	ሲ	ሳ	ሴ	ሰ	ሱ			ሲ		
š	ˆs	ሸ	ሹ	ሺ	ሻ	ሼ	ሸ	ሹ			ሺ		
q	q	ቀ	ቁ	ቂ	ቃ	ቄ	ቅ	ቆ	ቈ	ቀ	ቁ	ቂ	ቀ
q	.q	ቐ	ቑ	ቒ	ቓ	ቔ	ቕ	ቆ	ቈ	ቐ	ቑ	ቒ	ቐ
b	b	በ	ቡ	ቢ	ባ	ቤ	ብ	ቦ	ቡ	ቢ	ቢ	ቢ	ቡ
v	v	ቨ	ቩ	ቪ	ቫ	ቬ	ብ	ቦ			ቪ		
t	t	ተ	ቱ	ቲ	ታ	ቴ	ት	ቶ			ቲ		
č	ˆc	ቸ	ቹ	ቺ	ቻ	ቼ	ች	ቾ			ቺ		
h	_h	ኀ	ኁ	ኂ	ኃ	ኄ	ኅ	ኆ	ኸ	ኸ	ኂ	ኃ	ኸ
n	n	ነ	ኑ	ኒ	ና	ኔ	ኑ	ኖ			ኒ		
ñ	˜n	ኘ	ኙ	ኚ	ኝ	ኞ	ኙ	ኟ			ኚ		
,	,	አ	ሉ	ሊ	ላ	ሌ	ለ	ሉ	አ				
k	k	ከ	ኩ	ኪ	ካ	ኬ	ከ	ከ	ከ	ከ	ከ	ከ	ከ
k	_k	ኸ	ኹ	ኺ	ኻ	ኼ	ኸ	ኸ	ኸ	ኸ	ኸ	ኸ	ኸ
w	w	ወ	ዉ	ዐ	ዑ	ዒ	ወ	ዐ					
‘	‘	ዐ	ዑ	ዒ	ዓ	ዔ	ዐ	ዐ					
z	z	ዘ	ዙ	ዚ	ዛ	ዞ	ዘ	ዘ			ዘ		
ž	ˆz	ዠ	ዡ	ዢ	ዣ	ዤ	ዠ	ዡ			ዢ		
y	y	የ	ዩ	ዪ	ያ	ዬ	የ	ዮ	የ				
d	d	ደ	ዱ	ዲ	ዳ	ዴ	ደ	ደ			ዲ		
d	.d	ደ	ዱ	ዲ	ዳ	ዴ	ደ	ደ			ዲ		
g	ˆg	ጀ	ጀ	ጀ	ጀ	ጀ	ጀ	ጀ			ጀ		
g	g	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ
g	.g	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ	ገ
t	.t	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ			ጠ		
č	ˆC	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ			ጠ		
p	.p	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ			ጸ		
s	.s	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ	ጸ			ጸ		
ç	.c	ፀ	ፀ	ፀ	ፀ	ፀ	ፀ	ፀ			ፀ		
f	f	ፈ	ፉ	ፊ	ፋ	ፌ	ፋ	ፌ	ፈ	ፈ	ፈ	ፈ	ፈ
p	p	ፐ	ፑ	ፒ	ፓ	ፔ	ፐ	ፑ	ፐ	ፐ	ፐ	ፐ	ፐ
q	'q	ቐ	ቑ	ቒ	ቃ	ቄ	ቅ	ቆ					
k	'k	ኸ	ኹ	ኺ	ኻ	ኼ	ኸ	ኸ					
h	'h	ኸ	ኹ	ኺ	ኻ	ኼ	ኸ	ኸ					
g	'g	ገ	ገ	ገ	ገ	ገ	ገ	ገ					

~mA ሸጐ
~ri ሸ
~fi ሸ

Table 2: The Ethiopian characters

3.3 Punctuation

Although the punctuation characters look different from the punctuation of the Latin script, they have essentially the same meaning. We made some compromises between visual similarity and similar interpretation when we chose the encoding of the punctuation characters. In Tab. 3–5 we have collected the appropriate inputs for each of the characters.

⌘

input		:=	:-	::	,	;		:	:	?	'?	!	'!		...
output		÷	≈	≡	⌣	⌢		:	⌘	?	¿	!	¡		...

Table 3: The Ethiopian punctuation characters

input		<	<<	>	>>		,	,	,	,	,	,	,		...
output		<	«	>	»		,	”	”	”	”	”	”		...

Table 4: The Ethiopian quotation characters

input		(	)		[	]		\{	\}		\\$
output		(	)		[	]		{	}		\$

Table 5: Special characters

The punctuation characters match their SERA equivalents closely.

3.4 Spaces

When the Ethiopian script is printed today, an interword gap is signalled by a white space, in the same way as it is done for the Latin script. This kind of spacing can be used simply as in ordinary L^AT_EX documents.

But in former times word breaks used to be denoted by the character ⌘ and even today this method is used for handwriting. To get the appropriate effect look at the following L^AT_EX source

```
'abAs : 'alamanahē: 'abAs : 'alamanahē:
'abAs: 'alamanahē: 'abAs: 'alamanahē:
'abAs: 'alamanahē: 'abAs: 'alamanahē:
'abAs: 'alamanahē: 'abAs: 'alamanahē::
```

which results in

አባስ ፡ አለምነህ ፡ አባስ ፡ አለምነህ ፡ አባስ ፡ አለምነህ ፡ አባስ ፡ አለምነህ ፡ አባስ ፡
አለምነህ ፡ አባስ ፡ አለምነህ ፡ አባስ ፡ አለምነህ ፡ አባስ ፡ አለምነህ ፡

in the output. As you can see, line breaks are allowed after the ⌘ even if there is no space character in the source. Space characters immediately following or preceding a ⌘ in the input are ignored as we can see from the first line. Therefore newlines in the input will not cause any problem either.

3.5 Line breaking

The ordinary spaces as well as the white space surrounding an `:` can be stretched a little, so that it is possible to achieve proper justification. Nevertheless, the lack of hyphenation in the Ethiopian language makes itself felt from time to time, when $\text{\TeX}$ cannot find suitable breakpoints for a paragraph.

There are a few standard solutions to this problem, the easiest is to use a `sloppypar` environment which allows $\text{\TeX}$ to stretch the interword spaces more than usual. But this does not work when some words are simply too long. In that case one may want to rewrite the sentence that causes the bad break, maybe only by changing a few words.

However, the text might not be easily changable, e.g. because it is a quote from some other source, or the author insists on that very phrase. In that case you can insert a `\-` into a suitable breakpoint where the text will be split between two lines. Unlike the usual $\text{\TeX}$ behaviour, no hyphen will be added at the breakpoint. This kind of line breaking is especially well suited when the character `:` is used for interword spaces.

But maybe even the insertion of break points is impossible. In this case, some explicit `\hspace` must be added in a suitable position or a raggedright layout must be selected during the whole document or part thereof.

3.6 Numbers

Since today Arabic numbers are more frequently used than the original Ethiopian numbers, the `ethiop` package outputs the Arabic numbers when the letters 0 up to 9 occur in the source code.

But Ethiopian numbers can be typeset, too, by using the command `\ethnum`. If we enter `\ethnum{1}` we get አ as the result. But the macro `\ethnum` can do more than that. In fact it can convert all numbers up to 999 999 to their Ethiopian equivalents: `\ethnum{999999}` gives አአአአአአአአአአ. We can see that the program knows quite a lot about the representation of large numbers.

Arabic	1	2	3	4	5	6	7	8	9
Ethiopian	አ	አ	፩	፪	፫	፬	፭	፮	፯

Arabic	10	20	30	40	50	60	70	80	90	100	10000
Ethiopian	፲	፳	፴	፵	፶	፷	፸	፹	፺	፻	፳፻፲፱

Table 6: The Ethiopian numbers

In $\text{\LaTeX}$ we must output the contents of a counter from time to time. This can be accomplished using the macro `\ethiop`. Saying `\ethiop{subsection}` will cause $\text{\TeX}$ to print አ, since this is subsection 6.

3.7 Math mode

One of $\text{\TeX}$'s most important features is its math mode. We can use math within Ethiopian text, but by default all letters in mathematical formulas will be taken from the ordinary $\text{\TeX}$ fonts.

Ethiopian letters can be used in a formula, although this will require slightly more work. When the macro `\ethmath` appears in math mode while the Ethiopian language is selected, its single argument will appear in the proper size typeset with the Ethiopian fonts.

For example

```
$$$b+\frac{d}{f^g}\iff
\ethmath{ba}+\frac{\ethmath{da}}{\ethmath{fa}^{\ethmath{ga}}}}$$$
```

will result in

$$b + \frac{d}{f^g} \iff a + \frac{g}{d.7}$$

where you should note the varying fonts in the first and second subformula. Actually all the work is done by the macro `\ethmath` that outputs the Ethiopian characters.

3.8 Ethiopian dates

The Ethiopian calendar is based on the Julian calendar with twelve months of 30 days and one month of 5 days. Every fourth year is a leap year, which means that the last month will have 6 days. The calendar system is implemented in our package, so it is possible to type `\today` and get **ህዳር 28 1991**. (This is the date on which this document has been translated. Compare it to the date on the title page!)

3.9 Two examples

We provide the first sample text to illustrate the appearance of our font. The $\LaTeX$ source of the text begins with

```
\subsubsection*{'amala~nAytun mArq}

pAduwA bametbAl web ya'i.tAliyA katamA si~nor bAptisetA
yatasa~nu 'and tegu_h 'sarAta~nA yenofu nabar:: 'enih sawe
hulat qon^go sEto^c le^g^c nabaruA^caw::
```

and gives the following result:

በመለኛይቱን ማርቅ

ፓዱዋ በምትባል ውብ የኢጣሊያ ከተማ ሲኖር ባፕቲስታ የተሰኙ አንድ ትጉጎ ሠራተኛ ይኖሩ ነበር ። እኒህ ሰው ሁለት ቆንጆ ሴቶች ልጆች ነበሯቸው ። ኳዲሐሊና የተባለችው ውበቷና ጠባኛ የተሟላ ነው ። ፔፌሊ ውበቷ እንደ እጎቷ አሟልቶ ስላልሰጣት በመለኛነቷ ትታወቃለች ።

ፔፌሊ ላባቷ የበኩር ልጅ በመሆኗ በዚህ አለብላቢት ምላሷ ሳቢያ ምናልባት ባል አጥታ ቁም-ቀር እንዳትሆን አባቷ ስጋት አድርባቸው ነበር ።

የፈሩት አልቀረም ለቆንጆዋ ከቬርኖ እስከ ቊዳኑላ ሲጎርፍ ለመለኛዋ አጠገቧ የሚደርስ ሰው ጠፋ ።

The next example will clarify the usage of bold and slanted Ethiopian fonts. Italic characters are mapped to slanted characters. The font selection works just as in ordinary $\LaTeX$ with NFSS. The source text

```
'adis 'ababA \textbf{'adis 'ababA}
\textsl{'adis 'ababA \textbf{'adis 'ababA}}
```

gives us አዲስ አበባ አዲስ አበባ አዲስ አበባ አዲስ አበባ as the output.

3.10 Using **ethiop** with ArabTeX

Starting with version 3.6i, **babel** is now compatible with ArabTeX. But still there are some problems with the many active characters which are used by **ethiop**, so that the two packages do not work right away.

But it is possible to use ArabTeX with **ethiop** by including the special style **etharab.sty** after ArabTeX has been loaded. Some internal macros of ArabTeX are redefined, so this is not guaranteed to work with every version of ArabTeX, but it has been successfully used with version 3.06g3 of ArabTeX.

```
\documentclass{article}
\usepackage{arabtex}
\usepackage[english]{babel}
\usepackage{ethiop}
\usepackage{etharab}
\begin{document}
  \selectlanguage{english}
  The Arabic (<al.har_t>) and the Ethiopian script
  (\selectlanguage{ethiop}sel.tAnE\selectlanguage{english})
  may occur within one sentence.
\end{document}
```

We do not provide the output of the example, because this document is intended to be translatable even in the absence of ArabTeX. In fact, everything works as usual, Arabic text can be inserted using the **arabtext** environment or using <> pairs. However, these commands must not be used in arguments to other commands or in command definitions. If that is desired the complete commands or command definitions must be enclosed in a **noethiop** environment.

```
\begin{noethiop}
\section{<al.har_t> -- cultivation}
\end{noethiop}
```

provides an example. If Ethiopian characters are needed, too, then a little trick is in order.

```
\def\temptext{%
  \selectlanguage{ethiop}sel.tAnE\selectlanguage{english}}
\begin{noethiop}
\section{<al.har_t> -- cultivation -- \temptext}
\end{noethiop}
```

But there should be really, really few occasions for such ugly code.

4 Advanced topics

Although the usage of the `ethiop` package is not really difficult once one gets used to it, there are a few points to be aware of. We will highlight the internal structure of the package first, to make it more plausible why some problems just cannot be easily patched away.

4.1 Implementation notes

Because we have to deal with so many characters, we placed them in two separate fonts. The two codetables are shown in Tab. 7 and Tab. 8, together with the recommended input strings. (Remember that there might be different ways to achieve the same result.)

In the first font we preferred to place characters that result from a $\mathrm{T\!E\!X}$ ligature (which must not be confused with a ligature from ordinary printing) in the positions 0–31 and 128–255. These characters usually cannot be entered from a keyboard and hence it is safe to assume that they resulted from a ligature.

Only the characters in the primary font are accessible by entering ordinary characters and forming ligatures. For the other characters it is necessary to explicitly select the secondary font within the $\mathrm{T\!E\!X}$ code. But this requires the execution of $\mathrm{T\!E\!X}$ macros, hence the activation of some characters was required. The activated characters can inspect the following characters and request the necessary font change. In fact, this method of enlarging the number of available characters is quite general and might be used for other languages, too, e.g. to provide a unified input mechanism for all Latin characters.

For a complete documentation of the input parsing mechanism we refer the interested reader to the commented source code in `ethiop.dtx`, which can also be typeset by $\mathrm{L\!A\!T\!E\!X}$ to get a more readable version.

A long calculation is required to convert the Gregorian date provided by the $\mathrm{T\!E\!X}$ primitives `\year`, `\month`, and `\day` to the Ethiopian date. The implementation in $\mathrm{T\!E\!X}$ is straightforward, but hardly readable, because $\mathrm{T\!E\!X}$'s expressiveness for formulas is very weak. As calendar routines are provided for all the other `babel` language definitions, it was obvious that this problem had to be addressed.

4.2 Common pitfalls

After discussing some of the internals of the `ethiop` package, we are now prepared to examine some of the problems that result from our implementation.

First of all the characters `~`, `^`, `'`, `_`, and `.` are made active. This is unavoidable, but there are some drawbacks.

- We cannot use `^^` for entering special characters. Usually this is done in package files only, so we do not get into real trouble, since `babel` activates the characters only at the beginning of the document.
- We cannot use a `.` in numbers and $\mathrm{T\!E\!X}$ dimensions while Ethiopian text is being typeset. We can circumvent this problem by using a `,` instead of the `.` when entering numbers for $\mathrm{T\!E\!X}$. Note, that we can use the `.` without problems when we have temporarily switched to a language other than Ethiopian, e.g. English.

	0	1	2	3	4	5	6	7
0	mua ሙጦ	muI ሙጦ	muA ሚጦ	muE ሚጦ	mue ምጦ	qua ቁ	quI ቁ	quA ቁ
8	quE ቁ	que ቁ	bua ቡ	buI ቡ	buA ቢ	buE ቢ	bue ቡ	kua ኡ
16	kuI ኡ	kuA ኡ	kuE ኡ	kue ኡ	gua ጐ	guI ጐ	guA ጐ	guE ጐ
24	gue ጐ	fua ፈ	fuI ፈ	fuA ፈ	fuE ፈ	fue ፈ		
32		! !	’ ’ ”	’ ! ፤	\$ \$	’ ? ፈ		’ ’
40	(())		:= ÷		, ፣	- -	. .	/ /
48	0 0	1 1	2 2	3 3	4 4	5 5	6 6	7 7
56	8 8	9 9	: :	; ፤	< <		> >	? ?
64		‘A ግ				‘E ግ		
72		‘I ግ						‘O ግ
80						‘U ግ		
88				[[	‘ ‘ “	]]		
96	‘ ‘	‘a ግ	be ብ		de ድ	‘e ግ	fe ፍ	ge ግ
104	he ህ			ke ከ	le ል	me ም	ne ን	
112	pe ጥ	qe ቅ	re ር	se ስ	te ት		ve ቨ	we ው
120		ye ይ	ze ዝ	:- ፦	፡	: ፡፡	:: ፡፡	
128	ha ሀ	hU ሀ	hI ሂ	hA ሃ	hE ሄ	hO ህ	la ለ	lU ሉ
136	lI ሊ	lA ላ	lE ሊ	lO ሎ	ma ሙ	mU ሙ	mI ሚ	mA ሚ
144	mE ሚ	mO ሞ	ra ረ	rU ሩ	rI ረ	rA ራ	rE ሬ	rO ሮ
152	sa ሰ	sU ሰ	sI ሰ	sA ሰ	sE ሰ	sO ሰ	qa ቀ	qU ቀ
160	qI ቀ	qA ቃ	qE ቀ	qO ቀ	ba ብ	bU ቡ	bI ቢ	bA ቢ
168	bE ቢ	bO ቡ	va ቨ	vU ቨ	vI ቨ	vA ቨ	vE ቨ	vO ቨ
176	ta ተ	tU ተ	tI ተ	tA ተ	tE ተ	tO ተ	na ነ	nU ነ
184	nI ነ	nA ነ	nE ነ	nO ነ	ka ከ	kU ከ	kI ከ	kA ከ
192	kE ከ	kO ከ	wa ወ	wU ወ	wI ወ	wA ወ	wE ወ	wO ወ
200	za ዘ	zU ዘ	zI ዘ	zA ዘ	zE ዘ	zO ዘ	ya የ	yU የ
208	yI የ	yA የ	yE የ	yO የ	da ደ	dU ደ	dI ደ	dA ደ
216	dE ደ	dO ደ	ga ገ	gU ገ	gI ገ	gA ገ	gE ገ	gO ገ
224	fa ፈ	fU ፈ	fI ፈ	fA ፈ	fE ፈ	fO ፈ	pa ጥ	pU ጥ
232	pI ጥ	pA ጥ	pE ጥ	pO ጥ	pua ጥ	puI ጥ	puA ጥ	puE ጥ
240	pue ጥ						huA ዘ	vuA ሃ
248	tuA ተ	nuA ነ	zuA ዘ	yua የ	duA ደ	luA ሊ	ruA ረ	suA ሰ

Table 7: The Ethiopian codetable ETHA

	0	1	2	3	4	5	6	7
0	.qua ቁ		.quI ቁ	.quA ቁ	.quE ቁ	.que ቁ		
8	_hua ኀ		_huI ኀ	_huA ኀ	_huE ኀ	_hue ኀ		
16	_kua ኀ		_kuI ኀ	_kuA ኀ	_kuE ኀ	_kue ኀ		
24	.gua ኀ		.guI ኀ	.guA ኀ	.guE ኀ	.gue ኀ		
32								
40	~mA ሾ	~ri ሾ	~fi ሾ					
48		፳	፳	፳	፳	፳	፳	፳
56	፳	፳	፳	፳	፳	፳	፳	፳
64	፳	፳	፳	፳	፳			
72	.ha ሐ	.hU ሐ	.hI ሐ	.hA ሐ	.hE ሐ	.he ሐ	.hO ሐ	.huA ሐ
80	'sa ሠ	'sU ሠ	'sI ሠ	'sA ሠ	'sE ሠ	'se ሠ	'sO ሠ	'suA ሠ
88	^sa ሸ	^sU ሸ	^sI ሸ	^sA ሸ	^sE ሸ	^se ሸ	^sO ሸ	^suA ሸ
96	.qa ቀ	.qU ቀ	.qI ቀ	.qA ቀ	.qE ቀ	.qe ቀ	.qO ቀ	
104	^ca ቀ	^cU ቀ	^cI ቀ	^cA ቀ	^cE ቀ	^ce ቀ	^cO ቀ	^cuA ቀ
112	_ha ኀ	_hU ኀ	_hI ኀ	_hA ኀ	_hE ኀ	_he ኀ	_hO ኀ	
120	~na ኀ	~nU ኀ	~nI ኀ	~nA ኀ	~nE ኀ	~ne ኀ	~nO ኀ	~nuA ኀ
128	'a ለ	'U ለ	'I ለ	'A ለ	'E ለ	'e ለ	'O ለ	'ua ለ
136	_ka ኀ	_kU ኀ	_kI ኀ	_kA ኀ	_kE ኀ	_ke ኀ	_kO ኀ	
144	^za ኀ	^zU ኀ	^zI ኀ	^zA ኀ	^zE ኀ	^ze ኀ	^zO ኀ	^zuA ኀ
152	.da ደ	.dU ደ	.dI ደ	.dA ደ	.dE ደ	.de ደ	.dO ደ	.duA ደ
160	^ga ደ	^gU ደ	^gI ደ	^gA ደ	^gE ደ	^ge ደ	^gO ደ	^guA ደ
168	.ga ኀ	.gU ኀ	.gI ኀ	.gA ኀ	.gE ኀ	.ge ኀ	.gO ኀ	
176	.ta ጠ	.tU ጠ	.tI ጠ	.tA ጠ	.tE ጠ	.te ጠ	.tO ጠ	.tuA ጠ
184	^Ca ጠ	^CU ጠ	^CI ጠ	^CA ጠ	^CE ጠ	^Ce ጠ	^CO ጠ	^CuA ጠ
192	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ	ጠ
200	.pa ጸ	.pU ጸ	.pI ጸ	.pA ጸ	.pE ጸ	.pe ጸ	.pO ጸ	.puA ጸ
208	.sa ጸ	.sU ጸ	.sI ጸ	.sA ጸ	.sE ጸ	.se ጸ	.sO ጸ	.suA ጸ
216	.ca ፀ	.cU ፀ	.cI ፀ	.cA ፀ	.cE ፀ	.ce ፀ	.cO ፀ	
224	'qa ቀ	'qu ቀ	'qi ቀ	'qA ቀ	'qE ቀ	'qe ቀ	'qO ቀ	
232	'ka ሸ	'ku ሸ	'ki ሸ	'kA ሸ	'kE ሸ	'ke ሸ	'ko ሸ	
240	'ha ሸ	'hu ሸ	'hi ሸ	'hA ሸ	'hE ሸ	'he ሸ	'ho ሸ	
248	'ga ኀ	'gu ኀ	'gi ኀ	'gA ኀ	'gE ኀ	'ge ኀ	'go ኀ	

Table 8: The Ethiopian codetable ETHB

We could expect that the activation of `^` and `_` spoils TeX's math mode, but this is not the case. In fact math mode behaves just like before, with subscripts and superscripts in their proper position.

The complex calendrical calculation require the allocation of a large number of counters to hold the intermediate results.

- In connection with other counter intensive packages L^AT_EX may run out of counters when using the `ethiop` package. Maybe we can get rid of two or three of the counters in the next version, but it will still remain a problem.

This problem is actually due to the lack of temporary counters in L^AT_EX, which are not provided, even though this is done for all the other types of registers.

Since our package is not yet an integral part of the `babel` system, we have some other difficulties to overcome.

- The `ethiop` package cannot be loaded via an option to `babel`. Instead an explicit `\usepackage{ethiop}` has to be used.
- We do not know what the future will bring. While this package *might* work with future versions of `babel`, there is no guarantee that it will. So you should keep your old version of `babel` until you are sure it works with the `ethiop` package or until an updated version of `ethiop` is issued.

Typing errors will usually not result in an error message. Instead a black rectangle will appear in the output, if some illegal character is encountered.

- Because it is allowed to enter consonants without a trailing vowel, there are plenty of typos that simply result in the wrong letters being printed.

So look at your finished document carefully.

4.3 Support of the Ω typesetting system

The advantage of using Ω for typesetting the Ethiopian language is that it can handle text files encoded in Unicode. Therefore the complex, timeconsuming and error-prone conversion process needed our transliteration within TeX can be skipped.

Ω can still profit from `babel`'s support for captions, dates etc. In Ω you can simply enter the Ethiopian characters as their Unicode equivalents. They will be converted to our fonts by means of virtual fonts.

You will have to experiment when you want to use Ω with `ethiop`. Tests have been only rudimentary so far. There are some currently unused files that may help you. `ethotlit.otp` simulates the old transliteration process in Ω . `ethohyph.otp` allows word breaking between every syllable.

Let us note a few differences between our package and Unicode.

- The letter **ሂ** (**huA**) is not present in Unicode. It is mentioned in [16]. It is used in the language of Agew (**አገው**, also known as Awngi). [16] also uses **ኸ** (**_kuA**) as an alternative representation for the same sound.

An example word would be **ሶሂ**, which means in Agew *to eat*.

- The letter **ሙ** (**mui**) is equipped with a additional tail at the lower left in Unicode.

- The four series ፳ ('q), ፳ ('k), ፳ ('h), and ፳ ('g) are not present in Unicode. They are suggested as possible extensions, but will not occur in Unicode in the near future, as it seems.

5 Changes

Version 0.2 was the first version to be publicly released, but we will shortly list the changes that this version made to EthTeX.

- Several letters were added, most notably the Leslau extensions 'q, 'k, 'h, and 'g. Some diphthongs were added, too.
- The multi-letter encoding was chosen and implemented using **babel**.
- The calendar algorithms were programmed.

Version 0.3

- The letter ሀ (huA) was added.
- Some bugs regarding subscripts and superscripts in ordinary math mode were fixed.
- The command `\ethmath` was added to allow Ethiopian characters in math mode.
- Fonts in 5 point size were added to allow Ethiopian characters in subscripts and superscripts.

Version 0.4

- The punctuation characters ፡ and ። were added.
- The syllables ሙ (~mA), ሪ (~ri), and ፊ (~fi) were added. Now all characters of the proposed Unicode standard for the code positions U+0x1200h to U+0x137F are included.
- `\ethvariantCtrue` and `\ethvariantCfalse` were introduced. At the same time the shorthand `_C` was removed. The variant forms of the series ፡C can only be reached via the aforementioned commands from now on.

Version 0.5

- Problems with activating ' in math mode have been solved.

Version 0.6

- `etharab.sty` has been added to allow cooperation with ArabTeX.

Version 0.7

- As suggested by Donald E. Knuth, some faulty parameters in the files `etha8.mf` and `ethb8.mf` were corrected.
- As suggested by Donald E. Knuth, individual glyphs for the characters `ጸ` and `ጹ` were added. In previous versions these characters were combined from other glyphs.
- The `ethiop` package may now be loaded before or after `babel`, as desired. Previous versions required that `ethiop` had to be loaded after `babel`.
- Limited Ω support has been added.

6 Copyright and Liability Notice

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Although we do not require this as a part of our license, we would be very happy if you send us any changes you make.

7 Closing Remarks

Much work has been done during the development of this package, but much work had been done before and there are still open points.

7.1 To Do

What remains to be done?

- Bugs must be removed. We know that there are some, but we do not know which. Please report bugs to `ethiop@informatik.uni-hamburg.de`, we will be happy to make some updates.
- More languages must be added. To do this, we must get to know the names for ‘bibliography’, ‘chapter’, ‘index’ and so on for as many of the languages of Ethiopia as possible. Suggestions for these and other non-technical improvements can also be sent to `ethiop@informatik.uni-hamburg.de`.
- It must be possible to typeset whole books in the Ethiopian script. Although this is possible at the moment, there are several problems that occur with the different $\text{\LaTeX}$ structures like `\part`, `\section`, or even `\item`.
- We need to find out more about the conventions for typesetting in the Ethiopian script.

7.2 Thanks

We would like to thank Abass B. Alameneh, Johannes L. Braams, Donald E. Knuth, and Leslie Lamport for their efforts. The programs provided by them made our package possible.

Paul Seelig from the Debian team provided the necessary motivation for finally releasing **ethiop** under the GNU GPL.

Daniel Yacob gave encouraging and constructive feedback on our project. Michal Jerabek did some extensive field testing of our package.

While we are on the subject, here is a list containing some of the names that appeared in this text. You can already guess it, they are written in the Ethiopian script. But which name is which?

ብርሃኑ በየን	ዮሐንስ ብራምስ	ማንፍሬድ ኩድሊክ
አባስ አለምሃህ	አላፍ ኩመር	ለስሊ ለምፖርት
ዮሸን ምጽንገር	ዶናልድ ክኑት	ደንኤል ያዕቆብ

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